

Mymensingh Engineering College
1st Year 1st semester B. Sc in CSE
In course Examination- 2025
Subject: Chemistry, Subject code: 1104

Time: 1 Hour

Full Marks: 20

Answer any four questions of the following

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| 1. (a) What do you mean by ideal gas and real gas? | 2 |
| (b) Deduce an ideal gas equation ($PV=nRT$) from gas law. | 3 |
| 2. (a) What do you understand by chemical kinetics? | 2 |
| (b) Drive an expression for the rate constant of a first order reaction. | 3 |
| 3. (a) What is half life of a reaction? | 2 |
| (b) Explain that half life period of a second order reaction is related to initial concentration of the reaction. | 3 |
| 4. (a) What do you understand by reversible and irreversible reaction? | 2 |
| (b) Deduce the mathematical formula of the law of mass action. | 3 |
| 5. (a) What do you understand by K_p and K_c of a reaction? | 2 |
| (b) At 25°C temperature and 1 atm pressure PCl_5 is 80% dissociated into PCl_3 and Cl_2 respectively. Calculated the K_p and K_c . | 3 |
| 6. (a) What is meant by P^H of a solution? | 2 |
| (b) Explain the mechanism of constant P^H value of a buffer solution. | 3 |

$P \propto \frac{1}{V}$
 $P \propto T$
 $n \propto V$

CH_3COOH
 CH_3COOH